What is Protein?

- Protein is a macronutrient, meaning the body needs it in large amounts.
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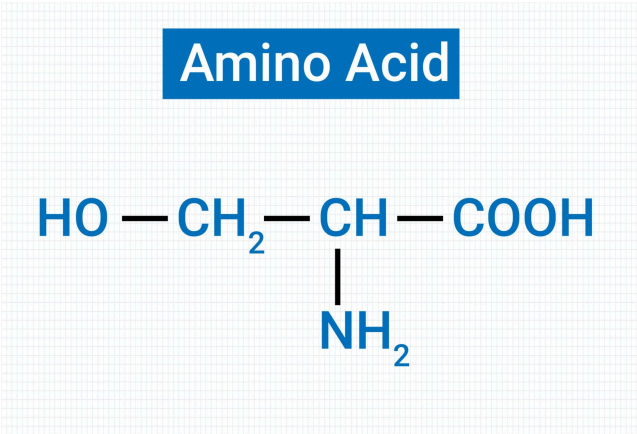
- It is made up of smaller building blocks called Amino Acids.
- The body contains thousands of different proteins, each with a specific job.

Main Functions (What it does)

Protein is not just for muscles; it does many things:

- Growth and Repair: It builds and fixes body tissues like skin, hair, and muscles.
- Enzymes: These are proteins that speed up chemical reactions in the body.
- Hormones: Some proteins act as messengers (like Insulin) to help body parts communicate.
- Immune System: Antibodies are proteins that help fight infections.
- Energy: While not the primary source, protein can provide energy if needed (4 kcal/g).
- Getty Images




Amino Acids: The Building Blocks

There are about 20 amino acids used by the body, divided into two groups:

1. Essential Amino Acids (9 total): The body cannot make these. You must get them from food.
2. Non-Essential Amino Acids (11 total): The body can make these on its own.

Protein Quality in Food

Not all protein sources are the same. They are categorized by which amino acids they contain:

1. High Biological Value (HBV) / Complete Proteins

- These contain all 9 essential amino acids.
- Sources: Mostly animal products like meat, fish, eggs, milk, and cheese.
- Plant exception: Soya and Quinoa are plant-based HBV proteins.

2. Low Biological Value (LBV) / Incomplete Proteins

- These are missing one or more essential amino acids.
- Sources: Mostly plant foods like beans, nuts, seeds, and grains.
- Protein Complementation: If you eat two different LBV foods together (like beans on toast), they provide all the essential amino acids your body needs.

Protein Requirements (How much do you need?)

- Average Adults: Need about 0.75g of protein per kilogram of body weight daily.
- Reference Intake: The general recommendation is roughly 50g per day.
- Who needs more?: Babies, children, and pregnant or breastfeeding women need extra protein for growth.

Deficiency and Excess

- Deficiency (Too little): Can lead to poor growth, thinning hair, and a weak immune system. In severe cases, it causes diseases like Kwashiorkor or Marasmus.
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- Excess (Too much): Very high protein intake can put stress on the liver and kidneys.

Would you like me to explain the specific "Protein Complementation" examples (like grains and pulses) in more detail?